

Identification and Classification of Encrypted Network Traffic Flows

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Topic

Analysis and comparison of Machine Learning (ML) tools for identification and classification of network traffic flows. Understanding the features used to propose the best methods to tackle security issues while studying encrypted data.

Security Challenge

The rapid shift toward end-to-end encryption (TLS 1.3, DoH) makes traditional Deep Packet Inspection (DPI) less effective. Security teams can no longer study the data inside packets to identify harmful activity directly. They use ML tools which require high-quality features. However, protocols like QUIC, TLS 1.3, and VPNs hide traditional identifiers.

The challenge is to determine which features better indicate malicious activity and which extraction tools provide the richest metadata to accurately classify traffic and detect threats without decryption and compromise of user privacy.

Proposed Approach

In this project we propose to compare specialized extraction frameworks like Joy, PcapPlusPlus, or NFStream in order to establish their effectiveness to identify encrypted flows of data.

We will try to :

- **Data Acquisition** : Generate web, streaming or VPN traffic.
- **Tool Comparison** : Use different tools on the same data set to extract statistical flows to determine interesting features like IAT, packet length, TLS fingerprints ...
- **Analysis** : Use more precise packet manipulation to find the most adapted features to detect threats and compare the result of the tools.

Relevant Tools

The implementation will rely on :

- **Extraction** : NFStream for Python integration, Joy for TLS/SSH fingerprints and PcapPlusPlus for C++ performance.
- **Analysis** : Python to compare outputs on known data.
- **Verification** : Pyshark and Scapy for packets manipulation.

Expected Results

The final output will be a comparison report of the different tools tested and their result on the features found to be determining. It will include :

- A comparative benchmark to show which tool provides the richest metadata for specific protocols.
- A model showing the ability of a tool to distinguish threats between different types of encrypted traffic using a confusion matrix to study accuracy.
- A feature roadmap defining a list of key features that indicate security threats while preserving privacy.